

LOST PLANE FINDER

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The lost plane finder introduced here is a little onboard beacon, also known as lost model alarm/crashed aircraft beacon, which helps you find a crashed or downed radio-controlled plane. The concept is to get an alert from a piezo sounder if there are no signals from the associated radio-controlled transmitter.

Usually, the radio transmitter provides a set of pulses every 20 milliseconds, which, in turn, sends the radio receiver a separate pulse to each of the servos at the same interval. In this circuit, an alarm will sound when the radio receiver no longer receives the set of pulses from the radio transmitter.

To locate the aircraft, all you have to do is switch your transmitter off and the system will start to beep. The circuit is very easy to build and does not require any special parts like costly microcontrollers. A typical lost plane finder is shown in Fig. 1.



Fig. 1: A typical lost plane finder

Circuit and working

Circuit diagram of the lost plane finder is shown in Fig. 2. It is built around timer NE555 (IC1), one transistor BC547 (T1) and a few other components. The design is centred around timer chip NE555 (IC1), configured here as a missing-pulse sensor. The whole circuit can be powered with power supplies ranging from 4V up to 9V, so you can use any power source that your radio receiver/servo setup can handle.

The 4-pin connector (CON2) is the input connector for the lost plane finder system, while the 2-pin connector (CON3) is an optional output connector. In the circuit, when the pulse input is available at input pin 4 of CON2, BC547 transistor (T1) is turned on and a high-level (H) voltage (near 2.6V when $V_{cc} = 4V$) is available through output pin 3 of NE555 (IC1). When the pulse input is absent (transmitter is off), T1 switches off and IC1 gives a low-level (L) output through its output pin 3.

As a result, the piezo buzzer (PZ1)

and the indicator (LED1) wake up to raise a noticeable alert. Presence of the optional output connector (CON3) enables driving of external low-voltage blinker/beeper loads, in case of an extreme demand for such enhancements.

Construction and testing

An actual-size, single-side PCB for the lost plane finder circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a plastic case.

A tiny perfboard is enough for the construction of the lost plane finder system. After construction, it is better to make the system vibration proof with the help of a heat-shrink tube, preferably a transparent type with appropriate diameter. Connect the finished system to any unused channel of your radio receiver through the input connector (CON2) as shown in the wiring diagram in Fig. 5.

If a free radio receiver channel is not available, the system can still be utilised by sharing an occupied channel with some other intermittently-used control like landing gear, flaps or airbrakes, with the help of a Y-Splitter servo cable attached to such a channel.

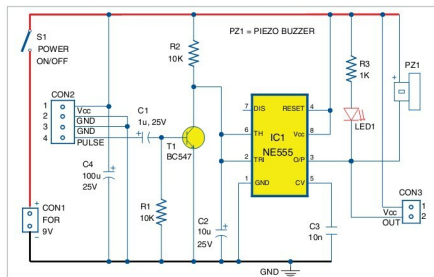


Fig. 2: Circuit diagram of the lost plane finder

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- T1 - BC547 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 10-kilo-ohm
- R3 - 1-kilo-ohm

Capacitors:

- C1 - 1 μ F, 25V electrolytic
- C2 - 10 μ F, 25V electrolytic
- C3 - 10nF ceramic disk
- C4 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 4-pin connector
- CON3 - 2-pin connector
- S1 - On/off switch
- PZ1 - Piezo buzzer
- 9V DC supply